

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

31st October 2014

Subject CT2 – Finance and Financial Reporting

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Recognition of any unpaid expenses at the end of the year in the financial statements is primarily the result of the..... concept.
- A)** Going concern
 - B)** Accrual
 - C)** Dual aspect
 - D)** Prudence
- [2]**
- Q. 2)** Free float is defined as
- A)** The proportion of equity of a company available for trading by the public on a stock market.
 - B)** The amount of money option traders need to deposit with the stock exchange as security.
 - C)** The amount of money a company keeps aside to meet any contingencies.
 - D)** The type of charge a company may create on its assets at the time of issuing bonds.
- [2]**
- Q. 3)** An imputed tax system ensures that
- A)** Individuals pay taxes at a higher rate as their income increases.
 - B)** An individual or company gets credit for taxes paid overseas on income earned abroad.
 - C)** Shareholders get tax relief on the dividend income as company has already paid tax on the profits earned.
 - D)** Indexation of purchase cost of the asset is allowed to be deducted from the sales price for the purpose of levy of capital gain tax.
- [2]**
- Q. 4)** Return on an asset as per CAPM is expected to be 11% while the risk free rate and the rate of return on the market portfolio are 5% and 10% respectively. If the standard deviation of the market return is 5%, the co-variance between the assets return and the market return would be
- A)** 6%
 - B)** 30%%
 - C)** 6%%
 - D)** 15%%
- [2]**
- Q. 5)** Which of the following are true about derivative contracts?
- I.** Derivative contracts are used only for arbitrage or hedging risks.
 - II.** Derivative contracts are traded either directly between two parties called Over-The-Counter (OTC contracts), or are traded via an exchange.
 - III.** Credit risk is a major factor in an exchange-traded contracts
 - IV.** Holding of capital and use of central clearing systems helps in reducing operational risk.
- A)** I, II, III & IV
 - B)** II & IV
 - C)** II only
 - D)** III & IV
- [2]**

- Q. 6)** Which of the following are regulated by a stock exchange?
- I.** The stock market
 - II.** Members of the stock exchange.
 - III.** Investors
 - IV.** Private companies
- A)** I, II, III & IV
B) I & II
C) I only
D) III & IV [2]
- Q. 7)** Which of the following are examples of Systematic Risks?
- I.** War between two oil producing countries
 - II.** Bankruptcy of a major multinational auto manufacturer
 - III.** Failure of a major power plant of the country
 - IV.** Floods affecting many industrial units in a city
- A)** None of the above
B) All of the above
C) I & III
D) II & IV [2]
- Q. 8)** According to Modigliani and Miller, which is the best capital structure for the company who wishes to maximise its market value (assuming no taxation)?
- a)** Company A
- | | |
|----------------------|------------|
| i. Equity | 15 Million |
| ii. Debt @ 8% | 15 Million |
- b)** Company B
- | | |
|----------------------|------------|
| i. Equity | 18 Million |
| ii. Debt @ 9% | 12 Million |
- A)** Company B
B) Company A
C) Both are equal
D) Insufficient information to decide on the best capital structure [2]
- Q. 9)** Which of the following can be classified as short term company finance?
- I.** Bank Overdraft
 - II.** Trade Credit
 - III.** Bills of exchange
 - IV.** Bank Loan
- A)** I, II & III
B) II & III
C) III only
D) I & IV [2]

Q. 10) Which of the following statements are true?

- I.** Legal ownership of asset passes to the buyer in case of a credit sale at the outset
- II.** Legal ownership of asset passes to the buyer in case of a hire purchase at the outset
- III.** Legal ownership of asset passes to the lessee in case of operating lease
- IV.** Legal ownership of asset passes to the lessee in case of finance lease

- A)** I, II & III
- B)** I & IV
- C)** II & III
- D)** I only

[2]

Q. 11) Arsenal Ltd, Mumbai has provided you with the following information with regards to its financial statements for the year ended on 31st March 2013 and 31st March 2014.

INR million

	As on 31st March 2013	As on 31st March 2014
Inventory	350.00	
Trade Payables		1200.00
Cash and Bank Balances	100.00	
Reserves and Retained earnings	6950.00	7900.00
Outstanding Expenses		300.00
Total depreciation charged on Office building and Machinery		1372.50
Manufacturing expenses other than Depreciation on machinery		1965.00

It has also provided you the information on the values of various financial ratios based on its financial statements for the year ended on 31st March 2014

Ratio	Value
Current ratio	2
Debtors turnover period	50 days
Creditors turnover period	36.5 days
Inventory turnover period	17.5 days
Interest cover	20 times
Asset utilisation ratio	1.9
Gross profit margin	23.15789%
ROCE (without considering Long term debt)	10%
Gross profit : Operating profit	4.4 : 1

Other Information Provided

- 1) Due to its export oriented activities, Arsenal Ltd is not liable to pay any taxes.
- 2) The company has two depreciable assets. It charges depreciation on these assets at the percentages mentioned against each asset on the value of the asset at the start of the year. No new purchases or disposals have occurred during the year ended on 31st March 2014
 - a) Building – 5% p.a.
 - b) Machinery – 15% p.a.
- 3) The company doesn't pay any dividends to its shareholders but rather uses the entire profits for expansion activities.
- 4) 100% of purchases of Raw material while 73% of the total Sales made during the year are on credit basis.
- 5) Arsenal Ltd holds some equity shares of Manchester United Ltd as investment.
- 6) The company has a Long Term Loan and interest on this loan is paid every year on 31st March
- 7) Round all calculated figures to 2 decimals.
- 8) The value of inventory used to calculate the Inventory Turnover Period is calculated as follows:

$$\text{Inventory used in calculations} = (\text{Inventory as at 31st March 2013} + \text{Inventory as at 31st March 2014}) \div 2$$
- 9) The company doesn't have any Current liabilities other than Trade payables and outstanding expenses. Its only Current assets are Inventory, Trade Receivables and Cash and bank balance.
- 10) For all calculations use 1 year = 365 days.

You are required to prepare the following in as much detail as possible:

- | | | |
|---------------|---|-------------|
| | i) Income statement for the year ended on 31 st March 2014. | (12) |
| | ii) Statement of financial position as on 31 st March 2014. | (10) |
| | | [22] |
| Q. 12) | i) Define depreciation. | (2) |
| | ii) State the two main methods of charging depreciation, and give the formulas to calculate the rate of depreciation. | (3) |
| | iii) Assuming that a particular asset needs to be completely depreciated over a fixed time period, which method of depreciation helps a company to report lower profits in the initial years? | (1) |
| | | [6] |
| Q. 13) | Your friend will be interviewed for the position of Finance Manager tomorrow. Amongst several other questions that he is preparing for, he seeks your help on the following three questions. Please advise him. | |
| | i) What is the primary objective of a Finance Manager? | (1) |
| | ii) What factors influence his decisions while trying to achieve his primary objective? | (2) |
| | iii) State any two factors that decide how an asset is financed? | (1) |
| | | [4] |

Q. 14) Mr Kohli wishes to invest in the equity of Anushka Ltd (a well-known company) after three months. He is expecting an inflow of \$195,950 from his overseas investments in a month's time. He has sought your advice on

i) The market risks to which he is exposed (1)

ii) The ways in which he could protect himself against the above risks. In case there are several alternatives available, please advise him on the key advantages and disadvantages of the alternatives (9)

[10]

Q. 15) Define the following terms

- Swap
- Warrant

[4]

Q. 16) What is an agency cost and when is it incurred?

[4]

Q. 17) i) Calculate the beta factor for the following investments.

Compare the returns on the investment to the CAPM expected returns and determine whether it is worthwhile to accept this investment at this level of risk.

Risk Free rate of interest can be taken at 6%

Probability	Market Returns	Investment Returns
1/3 (Recession)	9%	6%
1/3 (Normal)	12%	30%
1/3 (Boom)	18%	18%

(8)

ii) Will your answer be different if the probability of Recession is 2/3, Normal is 1/6 and Boom is 1/6.

(4)

[12]

Q. 18) A particular country has an uncertain investment environment. Each year could either be a Favourable scenario with 25% chance or could be Normal scenario with 40% chance or could be an adverse scenario with 35% chance.

A firm has an investment proposal requiring an outlay of Rs. 40,000 if favourable or normal scenario prevail in the beginning of the first year, however if the regulations change (adverse scenario in the first year) then the outlay could involve Rs. 50,000 in the beginning of the first year and Rs. 10,000 in the beginning of second year (irrespective of scenario in 2nd year).

The investment proposal is expected to have 2 years economic life with no salvage in normal or adverse scenario (however there could be a salvage of Rs. 5,000 at the end of the 2nd year if year 2 is a favourable scenario).

The cash inflow at the end of each year could be Rs. 20,000 in adverse scenario, or Rs. 25,000 in normal scenario or Rs. 30,000 in favourable scenario.

The firm uses 10% discount rate for this type of investment.

- i)** Construct a decision tree for the proposed investment project. Define the path and their joint probability. (12)
- ii)** What Net Present Value (NPV) will the project yield if worst outcome is realised? What is the probability of occurrence of this NPV? (2)
- iii)** What is the probability of getting a positive NPV? (2)
- iv)** Should the project be accepted? (2)

[18]
